Combined and Composite Games in Evoluationary Game Theory*

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Building upon previous work in analyzing finite population size effects on evolutionary games, defining new game as a compositon of well known 2 and 3 strategy games and derive replicator dynamics for different microscopic interaction processes. Computing drift and drift reversal (loss of stability below a critical population size) both through extensive simulations and analytical derivations for each of the agent interaction processes. Developed visualization tools.

Usage: Secondary publications and information retrieval purposes.

Structure: You may use the `description` environment to structure your abstract; use the optional argument of the `\item` command to give the category of each item.

I. INTRODUCTION: RPS DYNAMICS

To understand the combined game, first must look at the single cases of rock paper scissors and Snowdrift.

A. Rock-Paper-Scissors Model

Previous work focusses on the famous cyclical rock-paper-scissors game [1]

The general rock paper scissors matrix is defined as:

$$\begin{bmatrix} a & c & b \\ b & a & c \\ c & b & a \end{bmatrix} \tag{1}$$

For the standard symmetric case of the RPS game, $a = 0, c = -1, b = 1$ where rock crushes scissors, paper covers rock, and scissors cuts paper. Ties are awarded the payoff of 0 resulting in cyclical dynamics about the internal fixed point of $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$.

1. *Replicator dynamics*

Let x, y and z represent the fraction of players using rock, paper, and scissors respectively. z can be defined as $z = 1 - x - y$, reducing the dimension of the replicator equations. Expected payoffs for each strategy can then be defined as follows:

$$\pi_R = ax + b\left(-x - y + 1\right) + cy \tag{2}$$
$$\pi_P = ay + bx + c\left(-x - y + 1\right) \tag{3}$$
$$\pi_S = a\left(-x - y + 1\right) + by + cx \tag{4}$$

The replicator equations can be defined for (x, y) using the standard formula:

$$\dot{x}_i = x_i(\pi_i(x) - \langle \pi(x) \rangle) \tag{5}$$

where the average payoff (or mean fitness) in the population is:

$$\langle \pi(x) \rangle = x\pi_R + y\pi_P + z\pi_S$$

This leads to the following replicator equations for x and y:

$$\begin{aligned} \dot{x} = {} & xy\left(ax - ay - bx + b\left(-x - y + 1\right) + cy - c\left(-x - y + 1\right)\right) \\ & + x\left(-x - y + 1\right)\left(ax - a\left(-x - y + 1\right) - by + b\left(-x - y + 1\right) - cx + cy\right) \end{aligned} \tag{6}$$

$$\begin{aligned} \dot{y} = {} & xy\left(-ax + ay + bx - b\left(-x - y + 1\right) - cy + c\left(-x - y + 1\right)\right) \\ & + y\left(-x - y + 1\right)\left(ay - a\left(-x - y + 1\right) + bx - by - cx + c\left(-x - y + 1\right)\right) \end{aligned} \tag{7}$$

The stability of this system of replicator equations can be analyzed by computing the eigenvalues of the Jacobian

* A footnote to the article title

matrix of functions $F = \dot{x}, G = \dot{y}$.

Should include a figure for RPS cyclic dynamics.

Add in the fixed points and stability for RPS.

B. Snowdrift game

II. COMBINED GAME DEFINITION

This work focusses on the composite game with the idea of combing snowdrift (SD) and rock-paper-scissors (RPS) into a single four strategy payoff matrix. Average payoffs, replicator equations and fixed points can subsequently be derived and the deterministic dynamics can be studied. The simulation tool can do all of this, plot trajectories both in 2d and and in the 3d pyramid simplex.

Define our combined game and explain the subgame etc. Link to previous work about RPS

III. MODEL AND STOCHASTIC DYNAMICS

We consider a finite population of size N evolving under stochastic evolutionary dynamics with S strategies. The population state is described by the vector $\mathbf{x} = (x_1, \ldots, x_{S-1})$, where $x_i = n_i/N$ and the final strategy is fixed by normalization. At each update step, individuals interact according to a payoff matrix $\mathbf{A}$ and reproduce according to a specified microscopic update rule. In the large–N limit, the stochastic dynamics can be approximated by a Fokker–Planck equation governing the probability density $P(\mathbf{x}, t)$.

IV. FOKKER-PLANCK EQUATION

Master equation can be derived and using the 2nd order taylor expansion of the transition probabilities for each strategy, a fokker-planck equation can be derived for each of the processes.

In the diffusion approximation, the master equation may be expanded to second order in $1/N$, yielding a Fokker-Planck equation of the form

$$\frac{d}{dt}P^{\tau}(x) = \frac{1}{N}\left[-\frac{d}{dx}(a(x)P^{\tau}(x)) + \frac{1}{2}\frac{d^2}{dx^2}\left(b^2(x)P^{\tau}(x)\right)\right] \quad (8)$$

Here $x \in (i, j, k)$ denotes the frequency of a given strategy in the RPS subsystem, and

$$a(i) = T^{PR} + T^{SR} + T^{+R} - T^{RP} - T^{RS} - T^{R+},$$

$$b(i) = \sqrt{\frac{1}{N}\left(T^{PR} + T^{SR} + T^{+R} + T^{RP} + T^{RS} + T^{R+}\right)}$$

for strategy i, rock and similarly $a(j), b(j)$ for paper and $a(k), b(k)$ for scissors can be defined in the same way with the pairs of transition probabilities into and out of the respective states and must be derived for each microscopic update process and set of payoff matrices. Full derivation of Fokker-Planck equation and expressions for $a(x), b(x)$ can be found in full report detailing 2nd order expansion of the master equation for the 4x4 game.

For each strategy, $a(x)$ corresponds to the deterministic replicator equation ($N \to \infty$), and $b(x)$ is the diffusion term corresponding to the stochastic drift. Code is capable of computing these values for each interaction process in order to plot the correct determinisic adjusted dynamics. For the local update process, this $a(x)$ is equal to the standard replicator payoff difference equation (with some scaling factor accounting for selection pressure w).

V. DRIFT MEASURES

To characterize finite-population effects beyond deterministic dynamics, we study the expected one–step change of suitable coexistence measures,

$$\langle \Delta H \rangle = \mathbb{E}[H(\mathbf{x}_{t+1}) - H(\mathbf{x}_t)].$$

The sign of $\langle \Delta H \rangle$ determines whether stochastic drift favors interior coexistence or boundary attraction. $\langle \Delta H \rangle$ can be written as a sum over the simplex ($1 \leq i, j \leq N - i, k \leq N - i - j$).

$$\langle \Delta H \rangle = \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \left(H_s - H_{s'}\right) T^{s \to s'} \quad (9)$$

for all the states $s, s' \in \{R, P, S, L\}$ and transition probabilities $T^{s \to s'}$.

A. Rock-Paper-Scissors Drift

For the three-strategy subsystem, the replicator equations have the constant of motion:

$$H_{\text{RPS}} = -xyz \quad (10)$$

Also have the summation here.

$$\begin{aligned}\langle \Delta H_{\mathrm{RPS}} \rangle = \frac{2}{N}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\ \Big[z(y-x)(T^{RP}-T^{PR}) \\ +y(z-x)(T^{RS}-T^{SR}) \\ +x(y-z)(T^{SP}-T^{PS}) \\ +xz(T^{P+}-T^{+P})+yz(T^{R+}-T^{+R}) \\ +xy(T^{S+}-T^{+S})\Big] \\ +\frac{2}{N^2}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\ \Big[z(T^{RP}+T^{PR})+y(T^{RS}+T^{SR}) \\ +x(T^{SP}+T^{PS})\Big]\end{aligned} \tag{11}$$

This expression shows that finite population drift is controlled by antisymmetric differences of transition probabilities between competing strategies.

B. Snowdrift Drift

The 4th strategy occurs with frequency $q = 1-x-y-z$, we define the snowdrift coexistence measure

$$H_{\mathrm{SD}} = -q(1-q).$$

The expected value of H_{SD} over the entire simplex is then:

$$\begin{aligned}\langle \Delta H_{\mathrm{SD}} \rangle = \frac{12}{N}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\ \Big[q\big[(T^{R+}-T^{+R})+(T^{P+}-T^{+P}) \\ +(T^{S+}-T^{+S})\big]\Big] \\ +\frac{12}{N^2}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\ \Big[T^{+R}+T^{+P}+T^{+S}\Big]\end{aligned} \tag{12}$$

The leading $1/N$ term governs directional drift between the RPS subspace and the loner strategy, while the $1/N^2$ term represents finite–population diffusion.

C. Four-Strategy Coexistence Drift

To capture full four-strategy coexistence, the constant of motion H_4 is defined

$$H_4 = -xyzq,$$

which vanishes on all faces of the simplex and is maximized in the interior.

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N^7}\sum_{i=1}^{N}\sum_{j=1}^{N-i}\sum_{k=1}^{N-i-j} \Big[ijk(N-i-j-k)(T^{RP}+T^{RS}+T^{R+}+T^{PR}+T^{PS}+T^{P+} \\ +T^{SR}+T^{SP}+T^{S+}+T^{+R}+T^{+P}+T^{+S})-(i-1)(j+1)k(N-i-j-k)T^{RP} \\ -(i-1)j(k+1)(N-i-j-k)T^{RS}-(i-1)jk(N-i-j-k+1)T^{R+} \\ -(i+1)(j-1)k(N-i-j-k)T^{PR}-i(j-1)(k+1)(N-i-j-k)T^{PS} \\ -i(j-1)k(N-i-j-k+1)T^{P+}-(i+1)j(k-1)(N-i-j-k)T^{SR} \\ -i(j+1)(k-1)(N-i-j-k)T^{SP}-ij(k-1)(N-i-j-k+1)T^{S+} \\ -(i+1)jk(N-1-i-j-k)T^{+R}-i(j+1)k(N-1-i-j-k)T^{+P}-ij(k+1)(N-1-i-j-k)T^{+S}\Big]\end{aligned} \tag{13}$$

In the continuous limit

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[zq(y-x)(T^{RP}-T^{PR})+yq(z-x)(T^{RS}-T^{SR})+xq(y-z)(T^{SP}-T^{PS}) \\ +yz(q-x)(T^{R+}-T^{+R})+xz(q-y)(T^{P+}-T^{+P})+xy(q-z)(T^{S+}-T^{+S})\Big] \\ +\frac{6}{N^2}\int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[zq(T^{RP}+T^{PR})+yq(T^{RS}+T^{SR})+xq(T^{SP}+T^{PS}) \\ +yz(T^{R+}+T^{+R})+xz(T^{P+}+T^{+P})+xy(T^{S+}+T^{+S})\Big]\end{aligned} \tag{14}$$

This expression naturally decomposes into RPS–like drift terms, snowdrift–like drift terms, and diffusion cor-

rections. The sign of $\langle \Delta H_4 \rangle$ determines whether stochastic dynamics favors four–strategy coexistence or boundary attraction.

D. Comparison of drift measures

Drift reversal, the loss of stability of an attracting fixed point (something about hopf bifurcation) below a critical population size, occurs at the point where the value of $\langle H \rangle$ changes sign. Depending on the initial parameters, this effect is observed in the RPS and SD subgames, and in some cases in both.

Diagram of the main cases - large population attraction to center - one case where drift in SD but strong RPS giving the rod shape. Other case where strong SD but drift in RPS, giving the spiral / plane in the center of the simplex.

VI. INTERPRETATION

While $\langle \Delta H_{\mathrm{RPS}} \rangle$ and $\langle \Delta H_{\mathrm{SD}} \rangle$ describe drift within lower–dimensional subspaces, $\langle \Delta H_4 \rangle$ captures stability of full interior coexistence. Sign reversals of $\langle \Delta H_4 \rangle$ generally occur at larger population sizes, reflecting the additional geometric suppression of interior states in the four–strategy simplex.

[1] A. Traulsen, J. C. Claussen, and C. Hauert, Phys. Rev. Lett. **95**, 238701 (2005).